

SKILLUP × IBM • HACKATHON

CyberLens

AN AI-POWERED CYBERSECURITY AWARENESS
ASSISTANT FOR EVERYDAY USERS

Track: AI for Impact | Powered by IBM watsonx.ai & Streamlit

TEAM

DEXTURE

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Challenge

The Cybersecurity Awareness Gap

- ❖ **Rising Digital Threats**

Rapid online adoption is multiplying cyber attacks and deceptive scams globally.

- ❖ **The Human Element**

Phishing and social engineering tactics weaponize human trust rather than system bugs.

- ❖ **The Jargon Barrier**

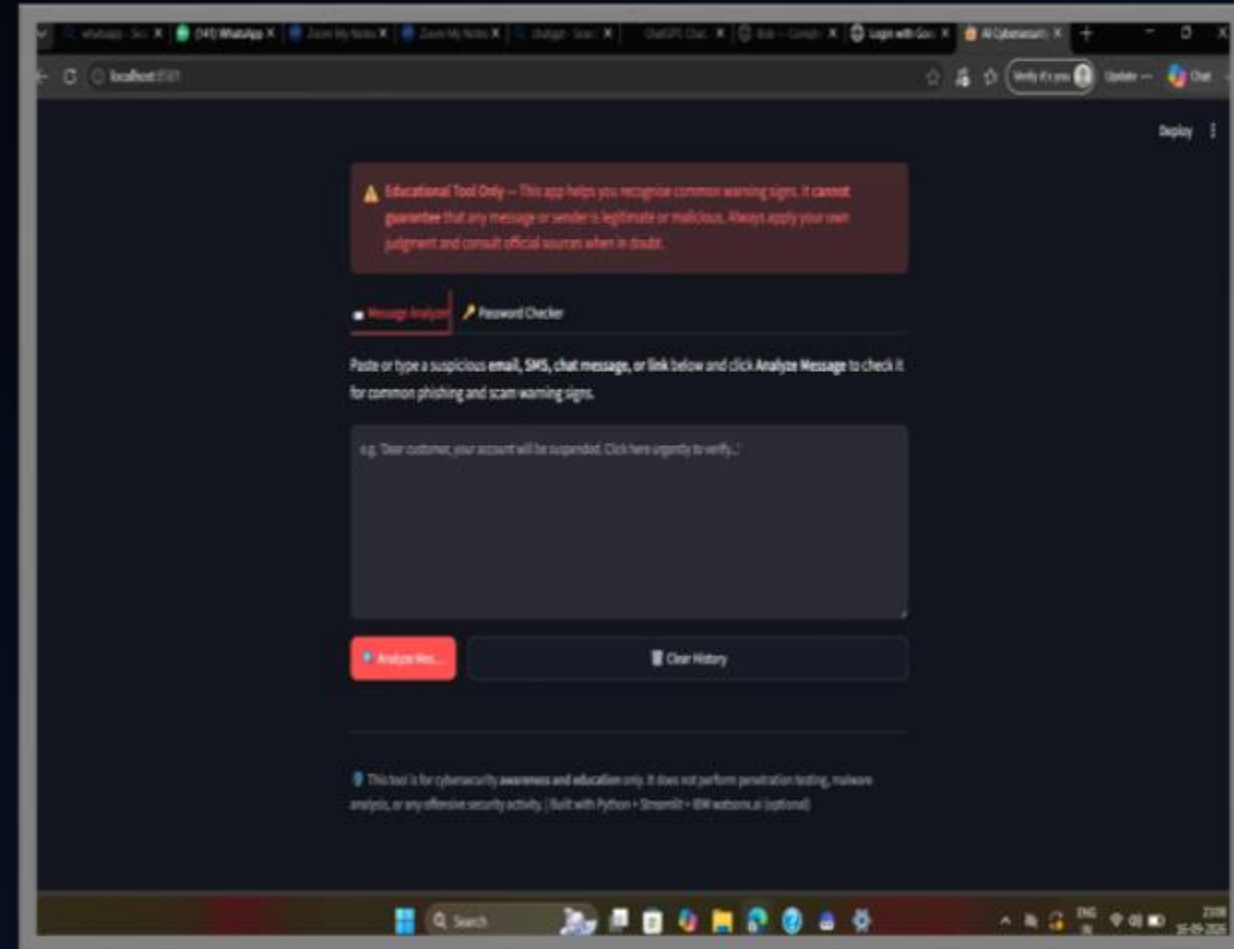
Traditional security warnings are overly complex, completely alienating everyday users.

Our Solution

CyberLens

Making Cybersecurity Awareness Simple

- ❖ **Analyze**
Detects phishing and scam warning signs in messages and links.
- ❖ **Assess**
Assigns a **Low / Medium / High** risk level.
- ❖ **Explain**
Uses **IBM watsonx.ai + Granite** to explain detected risks.
- ❖ **Guide**
Provides practical safety advice based on the risk.



CyberLens turns suspicious messages into clear risk insights—before users act.

How It Works

From **Input** to **Safety**

1

User Input

User enters an email, SMS, chat message, or link description.



2

Detect

Rule-based analyzer checks **7 warning indicators.**



3

Risk Score

The detected indicators are evaluated as:
LOW | MEDIUM | HIGH



4

Guide

CyberLens provides:

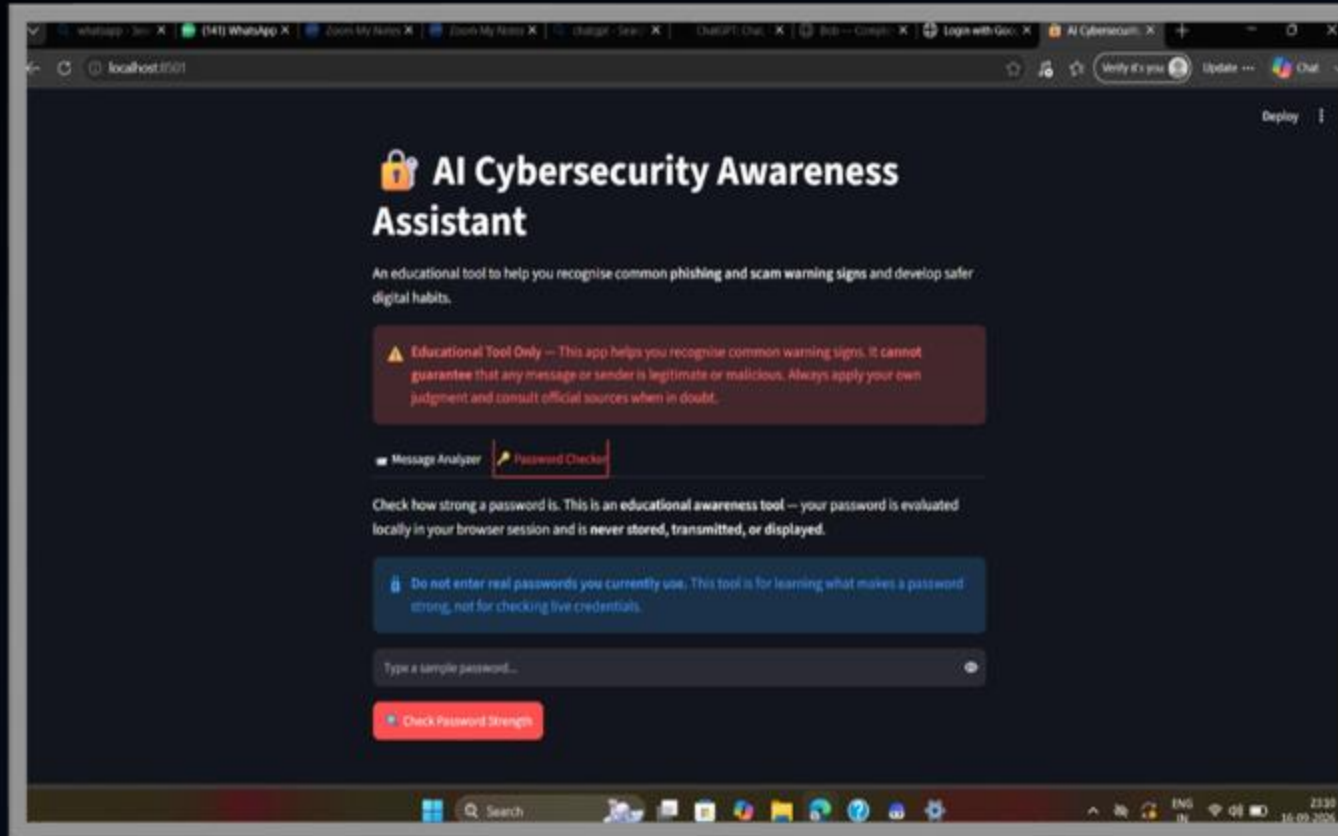
- Clear risk explanation
- Practical safety advice



IBM watsonx.ai + Granite
AI-generated, beginner-friendly explanation

Key Features

Built to turn cybersecurity awareness into everyday action.



1. Message Analyzer
Identifies common phishing and scam signs.

2. Password Checker
Provides educational feedback on password strength without storing the password.

3. Safety Guidance
Explains detected risks and suggests practical actions.

Simple tools. Clear explanations. **Safer decisions.**

AI + IBM

How AI Adds Value to CyberLens



Rule-Based Detection

Structured Detection:
CyberLens checks suspicious messages against **7 predefined warning indicators** and assigns a risk level.



IBM watsonx.ai + Granite

AI Explanation:
IBM watsonx.ai + Granite can turn the detected risks into **beginner-friendly explanations**.
Bottom



User Understands **The Risk**



Detection identifies the warning signs. AI helps users understand them.

Impact & Future Scope

Making Cybersecurity Awareness More Accessible



IMPACT

- ❖ Makes cybersecurity easier to understand
- ❖ Helps users recognize common scam patterns
- ❖ Encourages safer digital decisions



Future Scope

- ❖ Multilingual awareness support
- ❖ ML-based threat detection
- ❖ URL reputation analysis
- ❖ Expanded threat categories



From Awareness today → Stronger Protection Tomorrow



CyberLens

See The Threat, Know the Risk.
Stay Safe.



Analyze Suspicious
Messages



Understand the
Risk



Take Safe Actions

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